

Assignment-I

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Q1]

a) Simple returns are given by:

$$R(t) = \frac{(P_t - P_{t-1})}{P_{t-1}}$$

- 1st Month: $R(1) = \frac{108 - 100}{100} = 8\%$
- 2nd Month: $R(2) = \frac{103 - 108}{108} = -4.63\%$
- 3rd Month: $R(3) = \frac{115 - 103}{103} = 11.65\%$
- 4th Month: $R(4) = \frac{110 - 115}{115} = -4.35\%$

b) Log returns are given by:

$$r(t) = \ln \left[\frac{P_t}{P_{t-1}} \right]$$

- 1st Month: $r(1) = \ln \left(\frac{108}{100} \right) = 7.70\%$
- 2nd Month: $r(2) = \ln \left(\frac{103}{108} \right) = -4.73\%$
- 3rd month: $r(3) = \ln \left(\frac{115}{103} \right) = 11\%$
- 4th month: $r(4) = \ln \left(\frac{110}{115} \right) = -4.44\%$

c) Deviations: $|R_t - r_f| = D(t)$

$$D(1) = 0.3\%$$

$$D(2) = 0.1\%$$

$$D(3) = 0.65\% \rightarrow \text{Least accurate.}$$

$$D(4) = 0.09\%$$

↓

$$\text{Reason: } R(t) \approx r(t)$$

↓

Valid only for small returns.

↓

Longer returns $\Rightarrow$ longer deviations.

Q2]

a) Linearity of Expectations: $E(R_p)$

$\Rightarrow$ For random variables X & Y & constants a & b , the Expectation value of $aX + bY$ is given as.

$$E(aX + bY) = aE(X) + bE(Y)$$

↓

In portfolio returns, $R_p = w_1 R_1 + w_2 R_2$

$$\therefore E(w_1 R_1 + w_2 R_2) = E(R_p) = w_1 E(R_1) + w_2 E(R_2)$$

$$\therefore E(R_p) = 0.15(0.4) + (0.07)(0.6)$$

$$E(R_p) = 0.102 = 10.2\%$$

b) NO. The linearity of Expectations is independent of correlation b/w r.v.s.

Q3] $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \quad \text{--- (i)}$

$\text{Var}(aX) = a^2 \text{Var}(X) \quad \text{--- (ii)}$

$R_p = w_1 R_1 + w_2 R_2$

Let $X = w_1 R_1$ & $Y = w_2 R_2$

$\therefore \text{Var } X = w_1^2 \text{Var } R_1$
 $\therefore \text{Var } Y = w_2^2 \text{Var } R_2$ } From (ii).

$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y) \quad \text{--- (iii)}$

$\text{Cov}(w_1 R_1, w_2 R_2) = w_1 w_2 \text{Cov}(R_1, R_2)$

Putting all values in (i).

$\left[\begin{array}{l} \text{where, } \sigma_p^2 = \text{Var}(w_1 R_1 + w_2 R_2) \\ \sigma_1^2 = \text{Var}(R_1) \text{ \& } \sigma_2^2 = \text{Var}(R_2) \end{array} \right]$

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}(R_1, R_2)$

By correlation:

$\rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2}$

$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

Q4] $\sigma_A = 25\%$, $\sigma_B = 12\%$, $w_A = w_B = 0.5$

We know that $\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho \sigma_A \sigma_B$

• for $\rho = +1$,

$$\begin{aligned}\sigma_p^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho \sigma_A \sigma_B \\ &= 156.25 + 36 + 150 \\ &= 342.25\end{aligned}$$

$$\sigma_p = \sqrt{342.25} = 18.5\%$$

• for $\rho = 0$

$$\sigma_p^2 = 156.25 + 36 = 192.25$$

$$\sigma_p = 13.87\%$$

• for $\rho = -1$

$$\sigma_p^2 = 156.25 + 36 - 150 = 42.25$$

$$\sigma_p = 6.5\%$$

b) Wt. avg of individual risk: $w_A \sigma_A + w_B \sigma_B = 0.5(25) + 0.5(12)$
 $= 18.5\%$

$$\begin{aligned}\sigma_p^2 &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho \sigma_A \sigma_B \\ &= (\sigma_A w_A + \sigma_B w_B)^2\end{aligned}$$

OR.

$$\sigma_p = \sigma_A w_A + \sigma_B w_B = \text{Wt. avg. of individual risk.}$$

Hence proved.

c) For $\rho = -1$

$$\sigma_p^2 = (w_A \sigma_A - w_B \sigma_B)^2$$

$$\sigma_p = |w_A \sigma_A - w_B \sigma_B| \rightarrow 0$$

$$\therefore \frac{w_A}{w_B} = \frac{\sigma_B}{\sigma_A} = \frac{w_A}{1-w_A}$$

$$\therefore \boxed{(w_A + w_B) = 1}$$

$$\therefore 25 w_A = 12(1 - w_A)$$

$$37 w_A = 12$$

$$w_A = 32.43\%$$

$$w_B = 67.57\%$$

5] Weighted^{avg} risk sq: $(w_1 \sigma_1 + w_2 \sigma_2)^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2$

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Portfolio variance: $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$

Now, ρ_{12} is strictly less than 1

i.e. $\rho_{12} < 1$

$$\therefore \sigma_p^2 < (w_1 \sigma_1 + w_2 \sigma_2)^2$$

$$\therefore \sigma_p < (w_1 \sigma_1 + w_2 \sigma_2)$$

- The more we diversify, the less is the risk of loss by loss in a single asset as the other asset may cover up for it.

Q6]

a) $\Sigma_{ij} = P_{ij} \sigma_i \sigma_j$

$$\Sigma = \begin{bmatrix} 1\sigma_1^2 & (0.4)\sigma_1\sigma_2 & (0.1)(\sigma_1\sigma_3) \\ (0.4)\sigma_1\sigma_2 & 1\sigma_2^2 & (0.2)\sigma_2\sigma_3 \\ (0.1)\sigma_1\sigma_3 & (0.2)\sigma_2\sigma_3 & 1\sigma_3^2 \end{bmatrix} \quad \{ P_{ij} = P_{ji} \}$$

$$\Sigma = \begin{bmatrix} 625 & 120 & 10 \\ 120 & 144 & 9.6 \\ 10 & 9.6 & 16 \end{bmatrix}$$

b] In the upper matrix Σ

$\therefore \Sigma_{ij} = \Sigma_{ji}$ } the matrix is symmetrical.

Q7] Sharp Ratio: $S = \frac{E[R_p] - R_f}{\sigma_p}$

a)

• X: $S_x = \frac{14-4}{22} = 0.4545$

• Y: $S_y = \frac{9-4}{10} = \frac{5}{10} = 0.50$

• Z: $S_z = \frac{20-4}{35} = 0.4571$

b) $\therefore S(y) > S(z) > S(x)$,

Y earns 0.50% of excess return for every 1% of risk taken $\Rightarrow$ Rational Investor will prefer Portfolio Y.

c) Z has highest absolute return $\Rightarrow 20\%$

But

Z also has the highest extra risk $\Rightarrow 35\%$.

also $S_Z < S_Y$

$\Downarrow$

Z gives less reward per unit risk.

$\Downarrow$

A Rational investor might still prefer Y.

Q8]

a) The GMV portfolio is the portfolio with lowest possible risk among all wt. weight combinations.

- It is the leftmost pt. because no. portfolio can have lower σ than this.

$\Downarrow$

it is the leftmost pt. on horizontal axis.

b) NO.

Every portfolio on the lower half has an identical portfolio on the upper half with the same risk but higher return.

$\Downarrow$

Risk tolerance ~~can~~ ^{may} decide where you choose to be on the upper half $\Rightarrow$ NOT on the lower half

- c) CML is constructed by following steps.
- Plotting Efficient frontier.
 - Mark risk free asset at pt $(0, R_f)$.
 - Draw line from R_f tangent to efficient frontier

$$E(R_p) = R_f + \frac{E(R_M) - R_f}{\sigma_M} \cdot \sigma_p \quad \downarrow \text{CML}$$

Slope $\Rightarrow$ sharp ratio

M = max. risk adjusted return.

Q9]

- a) $n = 36$, $\sigma_i = 30\%$.

$\therefore$ Std error of sample mean

$$SE(\hat{\mu}_1) = \frac{\sigma_i}{\sqrt{n}} = \frac{30\%}{6} = 5\%.$$

$\downarrow$

Implication $\Rightarrow$ We cannot rely only on historical data alone while building portfolios. as the expected return may have error.

- b) MVO takes μ & Σ as inputs & finds the optimal wts. But Optimiser treats estimator μ to be accurate so if $\hat{\mu}$ is slightly wrong, ~~opt~~ error is amplified into the portfolio ~~est~~ weights.

- Replacing $\hat{\mu}$ with ML predicted returns helps because,
 - ML models see patterns beyond historical avg.
 - They produce more accurate & stable return estimates.

- c) $N = 50$, Σ is sym $\Rightarrow$ no. of unique = $\frac{N(N+1)}{2} = 1275$ parameters.

- Estimation of 1275 parameters is difficult & ~~difficult~~ ^{more assets we add, the more} difficult it becomes to predict.